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Tarcísio G. Rodrigues

Senior Software Engineer

Overview

Senior software engineer with 10+ years across C++, .NET and Java, building low-latency, high-reliability systems in fintech (market data, payments, risk analysis) as well as embedded software. Ramps up quickly on large, unfamiliar codebases, as shown by fixes merged upstream in open-source projects, and works directly with business stakeholders and third-party vendors. Currently applying spec-driven development with AI coding agents.

Employment

2026 **Software Engineer**, *Cetacea*, São Paulo, Remote

- Designed and developed .NET (C#) backend services for the customer ticket lifecycle, as well as middleware that interfaces with the Matrix communication service.
- Built and maintained a React/TypeScript/GraphQL frontend layer, adding a real-time widget to speed up customer onboarding.
- Led the team's adoption of spec-driven development with gsd-core: wrote the spec templates, defined the review workflow, and configured it to run out of the box under both Claude Code and OpenCode.

2025–2026 **Software Engineer**, *Mottu*, São Paulo, Hybrid

Built .NET (C#) backend services integrating with Santander's PIX payment infrastructure and BBVA México's payment services.

2024–2025 **Software Engineer**, *mParticle*, New York, Remote

Developed Java backend services, with some JavaScript frontend work, for a distributed query system that extracts marketing analytics (audience segmentation, funnel analysis) from large-scale customer event data.

2016–2023 **Software Engineer**, *XP Inc.*, São Paulo

- Built market data feed handlers for B3's UMDF platform in C++ (Boost.Asio, Signals2, Multi-Index, ICL), decoding FIX/FAST and SBE with QuickFIX, QuickFAST and the SBE reference implementations.
- Designed the infrastructure connecting backend and frontend services, using .NET micro-services, Kafka, MongoDB and WebSockets, deployed on Docker and Kubernetes.

2014–2016 **Software Engineer**, *FIT – Institute of Technology*, Sorocaba

Developed a C++ library for payment applications (EMV standard, Verifone SDK), with CMake and Python build automation.

2014 **Software Engineer**, *MAPS – Financial Market Solutions*, São Paulo

Developed risk analysis software in C++/MFC and maintained Jenkins CI.

2013–2014 **Software Engineer**, *IGS – International Gaming Solutions*, São Paulo

Built an embedded C++/Boost backend for bingo machines and its Buildbot CI, and ported legacy C code to Linux.

2012–2013 **Software Analyst**, *FiberWork – Optical Communications*, Campinas

Developed Android apps and embedded Java software for optical network test units.

Projects and open source

2026 **Author**, *carimbo*, LLM extraction of Brazilian invoices, .NET and Python

- Built a pipeline that uses Claude to extract NF-e invoices from DANFE PDFs into schema-validated records, checked by deterministic validators (check digits, access key vs. fields, tax arithmetic) and a bounded repair loop that flags errors without leaking the correct values.
- Measured it with a reproducible eval suite and per-run cost tracking; in live runs, validators turned misread access keys from silent successes into flagged failures.

2025–2026 **Contributor**, *devenv (Cachix)*, Nix-based developer environments, Rust and Nix

- Added Claude Code options for the primary agent and per-agent effort, with validation, tests and docs (#3069).
- Fixed a WSL2 interactive shell crash caused by a transient 0x0 terminal size (#3158).
- Added a treefmt integration (#2238) and an `allowUnsupportedSystem` option (#2639).

2026 **Contributor**, *FactorSeal (Cachix)*, hardware-backed secrets vault, Rust

- Fixed reliability bugs in the browser extension, approval flow and SecretSpec integration, such as reconnecting after a lost native host or session (12-commit PR #41).

2026 **Contributor**, *gsd-core*, spec-driven development framework for AI coding agents

- Diagnosed two bugs in `/gsd-pr-branch` (#4605, #4606); the fixes were incorporated into the upstream resolution (#4986).
- Root-caused and fixed a CI workflow that had failed on every scheduled run (#5035).

Education

2008–2010 **MSc in Philosophy**, *University of Campinas (Unicamp)*, Campinas
Mathematical Logic, with a dissertation on logic programming.

2002–2007 **BEng in Computer Engineering**, *Unicamp*, Campinas

Technical highlights

Programming	C++, C#/.NET, Java, Rust, Python, TypeScript/Node.js
Backend	ASP.NET Core, gRPC, REST, GraphQL, Boost/Asio, OAuth, JWT
Performance	Kafka, RabbitMQ, Redis Pub/Sub, LMAX Disruptor, io_uring, libuv
Data	PostgreSQL, SQL Server, MongoDB, Redis
DevOps	Docker, Kubernetes, ArgoCD, Nix/NixOS, GitHub Actions, Jenkins, GCP, AWS; Datadog, Grafana, Prometheus
Fintech	PIX payment APIs, FIX/FAST/SBE, B3's UMDF platform, EMV
AI tooling	Claude Code, OpenCode, spec-driven development (gsd-core), prompt and context engineering
Practices	DDD, CQRS, TDD, Clean Code, Scrum, Kanban

Languages

English (advanced), Portuguese (native), Spanish (conversational)